

Modeling the Effects of Bodyweight on Deadlift and Squat Performance Across Age and Sex

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Abstract

Powerlifting performance is influenced by an athlete's physical attributes, but the extent to which these factors affect performance is unclear. This study analyzes data from Open Powerlifting competitions to investigate how deadlift and squat performance vary across sex, age groups (Youth, Young Adult, Middle Age, Senior), and bodyweight. Using log-log regression, we find that bodyweight increases performance in both lifts. However, the strength of this relationship is sex and age dependent. A doubling in bodyweight results in a 2.28% greater increase in deadlift performance for males compared to females. In the squat, seniors experience a 5.23% smaller performance gain from a doubling in bodyweight than middle-aged lifters. Overall, the impact of bodyweight on both lifts is greatest for young adult males. In contrast, the effect is weakest for youth lifters in the deadlift and for seniors in the squat. We also find that bodyweight has a greater influence on squat performance than on deadlift performance across sex and age groups. These findings highlight the importance of considering interaction effects when modeling strength performance and have practical implications for training and weight class decisions in competitive powerlifting.

Introduction

This study analyzes the relationship between an athlete's physical attributes and their powerlifting performance, with a focus on the deadlift and squat. Specifically, we investigate how performance in these two lifts is related to sex, age class, and bodyweight. Additionally, we examine whether the effect of bodyweight on performance varies across age groups and between sexes, testing for an interaction between the variables. By comparing both squat and deadlift outcomes, we assess whether bodyweight plays a more significant role in influencing one lift over the other.

Understanding these interactions is valuable because it offers insight into how strength gains from weight gain may depend on an athlete's age and sex and event. This has practical implications for training and competition strategy, particularly when deciding whether moving up a weight class is likely to yield meaningful performance improvements. Given the strategic nature of powerlifting, our analysis aims to support more informed, individualized decisions for athletes of all ages pursuing long-term success in the sport.

The data that we used in this study came from the Open Powerlifting project, a community

driven initiative aimed at building a comprehensive, publicly accessible archive of global powerlifting competition results (Open Powerlifting). Our dataset was collected from the publicly available Open Powerlifting database, which consists of competition results from around the world. The data was compiled in a TidyTuesday project, dated October 8, 2019, publicly available on GitHub. The dataset includes detailed records for 17,803 unique lifters from over 70 countries across regions such as North America, Europe, Asia, and Oceania. Data collection began in the early 2000s and continues to the present day. Results are from competitions hosted by the International Powerlifting Federation (IPF), USA Powerlifting (USAPL), and various national and regional organizations. For each lifter, information such as sex, age, bodyweight, and performance outcomes for the squat, bench press, and deadlift are recorded.

Results

When analyzing the relationship between deadlift and squat performance (in kg) and sex, age class, and bodyweight (in kg), we began by plotting four graphs: two displaying performance versus bodyweight divided by age class, and another two divided by sex. The residual plots from the initial linear models indicated violations of linear regression assumptions, with clear signs of heteroskedasticity and curvature in both models.

To address these issues, we explored various transformations and found that a log-log transformation (i.e., $\log(\text{performance})$ vs. $\log(\text{bodyweight})$) produced the most linear and homoscedastic relationships for both deadlift and squat models. As shown in Figures 1-4, the log-log models revealed a positive, moderate, linear relationship between $\log(\text{performance})$ and $\log(\text{bodyweight})$ across age class and sex.

While the log-log transformation substantially reduced heteroskedasticity, the residual plots from the transformed models still exhibited minor signs of heteroskedasticity. Additionally, the Q-Q plots for both models showed slight left skewness. These deviations may affect the accuracy of standard errors and p-values, though the overall fit of the models was improved.

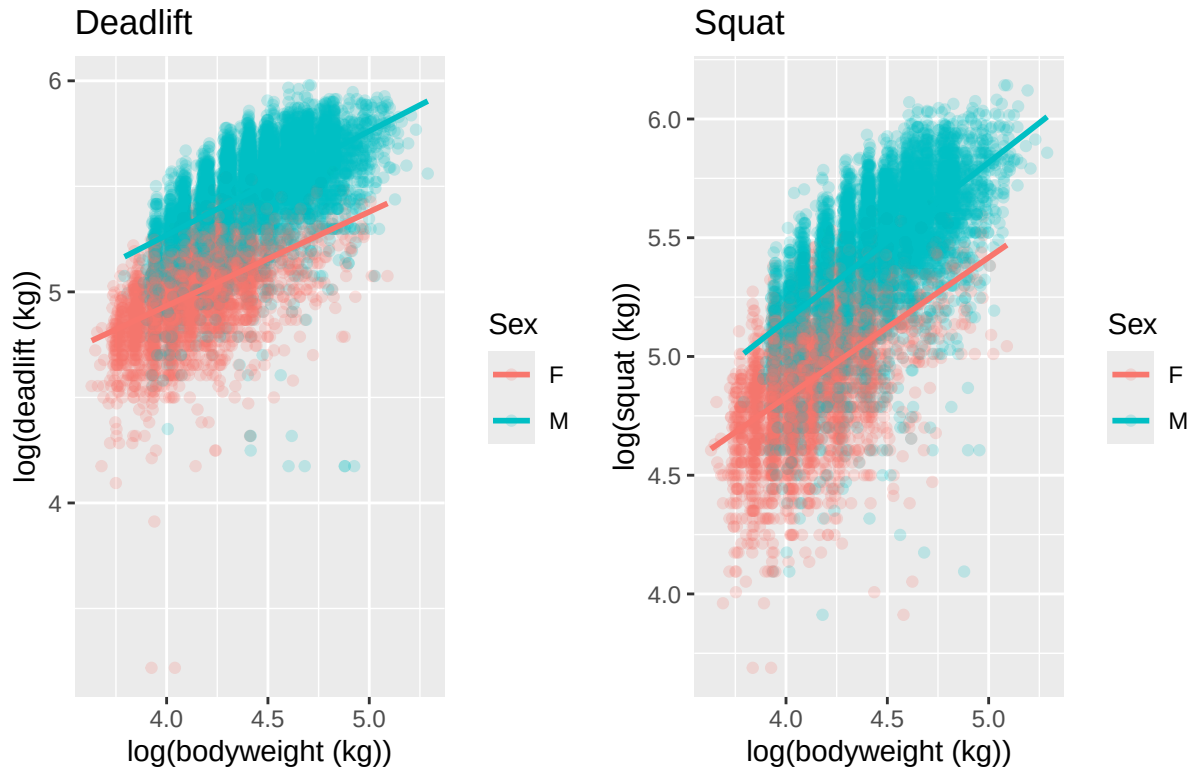
Figure 1 shows how lifting performance changes with bodyweight for different age groups: Youth, Young Adult, Middle Age. On the left plot of Figure 1, we can see that as bodyweight increases, deadlift performance also increases across all age groups. The Young Adult group shows the steepest line, meaning their performance improves the most with higher bodyweight. Seniors and Youth have shallower lines, showing less improvement. Similar to deadlift, if we see on the right plot of figure 1, the squat performance improves with bodyweight. The Middle Age and Young Adult groups show steeper lines, while Seniors show the least improvement.

Figure 1: Bodyweight and Lifting Performance by Age



Figure 2 shows the relationship between bodyweight and lifting performance for different sex. we can see on the left plot that both males and females show improved performance with more bodyweight, but the male line is steeper and higher. This means males tend to lift more at the same bodyweight and gain more strength when the weight increases. On the right plot, we can see that the pattern is similar. Males generally lift more than females and show greater performance gains as bodyweight increases. It is more steeper than the one for the deadlift.

Figure 2: Bodyweight and Lifting Performance by Sex



We analyzed outliers in both of our models through plots of the cooks distance, leverage and studentized residuals and noticed no influential points in the plots of cook's distance and leverage for both models. However, there were a few points below the threshold of -3 in the graph of the studentized residuals for both models. Despite this, there is no major influence from a single observation. Therefore, we decided against removing the observations with flagged studentized residuals.

To assess whether bodyweight interacts with age class and sex in predicting deadlift and squat performance, we conducted three ANOVA F-tests for each model. The first compared a model with only a bodyweight-by-age class interaction to the full interaction model. The second compared a model with only a bodyweight-by-sex interaction to the full model. The third compared a model with no interaction terms to the full model. For deadlift, all three tests yielded p-values below 0.05, indicating that including interaction terms significantly improves model fit and that the effect of bodyweight on deadlift performance depends on both age class and sex. For squat, however, the p-value for the model with only a bodyweight-by-sex interaction was 0.095, suggesting insufficient evidence to conclude that squat performance depends on age class.

Although the summary table for both full interaction models show that some individual interaction terms (e.g., between bodyweight and specific age groups) are not statistically significant, this does not rule out the presence of an interaction. Individual z-tests test whether each coefficient differs from zero, but they do not assess the joint significance of the interaction terms. The ANOVA F-test evaluates whether at least one interaction coefficient

is non-zero, providing evidence of a meaningful interaction effect overall.

To further support the use of an the full interaction models we performed Akaike Information Criterion (AIC) tests. For both deadlift and squat, the full models had the lowest AIC values, indicating a better fit relative to the simpler nested models. Despite the 0.095 p-value in one ANOVA F-test for the squat model, we retained the full interaction model, as the AIC favored it and the p-value was only slightly above 0.05. These findings justify the inclusion of interaction terms in both models.

The model for the deadlift is:

$$\begin{aligned} \log(\text{Deadlift}) = & \beta_0 + \beta_1(\log(\text{Bodyweight})) + \beta_2(\text{Male}) + \beta_3(\text{Senior}) + \beta_4(\text{YoungAdult}) \\ & + \beta_5(\text{Youth}) + \beta_6(\log(\text{Bodyweight}), \text{Male}) + \beta_7(\log(\text{Bodyweight}), \text{Senior}) + \\ & \beta_8(\log(\text{Bodyweight}), \text{YoungAdult}) + \beta_9(\log(\text{Bodyweight}), \text{Youth}) \end{aligned}$$

The unlogged model is:

$$\begin{aligned} \text{Deadlift} = & e^{\beta_0 + \beta_2(\text{Male}) + \beta_3(\text{Senior}) + \beta_4(\text{YoungAdult}) + \beta_5(\text{Youth})} \\ & * \text{Bodyweight}^{\beta_6(\text{Male}) + \beta_7(\text{Senior}) + \beta_8(\text{YoungAdult}) + \beta_9(\text{Youth})} \end{aligned}$$

The model for squat is the same as above except that the response variable is log(Squat) instead of log(Deadlift).

In order to address how the effect of bodyweight on deadlift and squat performance varies by age group and sex, we analyzed the coefficients of the interaction terms in the log-log regression models.

The table shows how doubling bodyweight affects deadlift/squat performance across sex and age groups, using coefficients from a log-log regression models. The interpretation of each coefficient in the table is as follows:

For *Group*, a doubling in bodyweight is associated with a *Event_Percentage* percent increase in deadlift/squat performance.

A tibble: 8 x 5

Group	Deadlift_exp	Deadlift_Percentage	Squat_exp	Squat_Percentage
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 Female, Mid-age	0.433	35.0	0.616	53.2
2 Female, Youth	0.403	32.3	0.547	46.1
3 Female, Young Adu~	0.457	37.2	0.600	51.6
4 Female, Senior	0.428	34.5	0.539	45.3
5 Male, Mid-age	0.465	38.1	0.670	59.1
6 Male, Youth	0.436	35.3	0.602	51.7

7 Male, Young Adult	0.489	40.4	0.654	57.4
8 Male, Senior	0.460	37.6	0.593	50.8

Overall, male young adults see the highest gain in deadlift performance for a doubling in bodyweight at 40.36%, while female youths have the lowest at 32.26%. Overall, males and young adults benefit the most in deadlift performance from added bodyweight, but the effect is consistently positive across all groups. These patterns are similarly reflected in squat performance, but with consistently greater percent increases. This suggests that bodyweight has a stronger multiplicative effect on squat performance than on deadlift performance across all demographic groups. For squat, male middle-aged lifters see the highest gain in performance from doubling their bodyweight at 59.80%, while female seniors have the lowest at 44.63%. The general trend is that, males and middle-aged individuals see greater proportional increases in squat performance when bodyweight doubles. However, in squat seniors rather than youth athletes see the least proportional increase in squat performance as bodyweight doubles.

Examining the interaction terms of the deadlift and squat regression models separately provides further insights. For this comparison, we put the results in the following table:

A doubling in bodyweight is associated with a *Event_Percent* percent increase for *Group 1* versus *Group 2* in deadlift/squat performance, holding all other variables constant.

A tibble: 4 x 3

Group	DL_Percent	SQ_Percent
<chr>	<dbl>	<dbl>
1 Male vs Female	2.28	3.85
2 Senior vs Mid-age	-0.343	-5.20
3 Young Adult vs Mid-age	1.65	-1.07
4 Youth vs Mid-age	-2.03	-4.63

Discussion

The ANOVA tests indicate that the effect of bodyweight on deadlift and squat performance varies across both age group and sex. Based on the interpretation of coefficients from the log-log models, we find that performance in both lifts increases with bodyweight across all groups. Holding age and bodyweight constant, males tend to outperform females in the deadlift and squat.

For the deadlift, the impact of bodyweight is strongest among young adult lifters (ages 18–39), particularly for males. In contrast, the effect is weakest in the youth category (ages 5–17), regardless of gender. These findings are supported by our graphical analysis, which shows the steepest slope between bodyweight and deadlift for young adults and the flattest for youth athletes, as well a steeper slope for males than females.

For the squat, the pattern is similar: the impact of bodyweight is also strongest among young adults, especially males. However, unlike the deadlift, the effect of bodyweight is weakest for seniors (age 65+), rather than youths. Graphs of squat performance versus bodyweight supports this, with the steepest slope for young adults and the flattest for seniors.

These results highlight that male lifters and athletes in the 18-39 age range gain more from increases in bodyweight in both lifts compared to very young, senior or female lifters. When comparing the two lifts, a doubling in bodyweight has a slightly greater effect on squat performance than deadlift performance. In addition, the difference in the effect of bodyweight between women and men is greater for squat than for deadlift.

From a physiological point of view, these findings align with established literature that attributes sex differences in strength to hormonal, neuromuscular, and muscle composition differences.. The findings show the importance of considering interaction effects in strength modeling and can inform training programs by emphasizing individualized performance expectations based on sex, age, and body composition.

Due to left-skewed behavior in the q-q plot of our log-log model, our data analysis can not be used to make predictions about an athlete's predicted deadlift or squat based on their gender, body weight and age class. In addition, the residual plot is still slightly heteroskedastic which means that the standard error of the coefficient estimates may be too small. This affected our ability to take accurate confidence intervals of the coefficients.

With additional time, we would have liked to explore several areas of analysis. First we would compare the effects of age class, sex and bodyweight on the Wilks score of powerlifting athletes. The Wilks score is a coefficient that takes into account the total weight lifted by a powerlifter across the three main lifts: squat, bench and deadlift. This metric controls for body weight and allows for a fairer comparison across lifters of different sizes, which could reveal whether performance disparities persist after accounting for body mass.

Second, we would expand on our analysis to include the bench press and total competition performance, offering a more complete picture of athletic strength and performance potential. Finally, we would address the remaining heteroskedasticity in our residual plots by using approaches beyond transformations, considering techniques beyond the scope of this course. These changes would help create a more reliable predictive tool for calculating an athletes bench or deadlift.

While our data comes from a publicly available dataset the collection and implications of our findings are important to consider. Using the American Statistical Association's Ethical Guidelines as a framework, we recognize our responsibility to uphold professional integrity and accountability throughout this analysis.

The ASA guidelines state that in data collection and handling, one should "protect people's privacy and the confidentiality of data concerning them." Although the competitions are public and we have removed individuals' names from our analysis, it is still important to consider how we are using the data as there is always a risk of identification or misuse. Ethical statistical practice requires that we remain thoughtful about preserving confidentiality, even when working with publicly accessible data.

More relevant to our analysis, the ASA also urges practitioners to "consider the impact of statistical practice on society, groups, and individuals," and to "recognize that statistical practice could adversely affect groups or the public perception of groups, including marginalized groups." In our study, we analyze performance differences across sex and age

categories. While these variables are central to our research question, there is a risk that certain interpretations, such as findings that male lifters outperform female lifters, or that older athletes respond differently to changes in body weight, could reinforce harmful stereotypes of marginalized groups, including women and senior lifters. These findings could potentially be misused to justify biased or exclusionary practices in competitive powerlifting. It is therefore crucial that we present and frame our results responsibly.

R CODE Appendix and References

```
#remove repeated data so that each competitor appears only once
data <- read.csv("ipf_lifts.csv")
cleaned_df <- data[!duplicated(data$name),]
```

```
#reducing the age category:
```

```
library(dplyr)
```

```
df_noempty <- df_noempty %>%
  mutate(age_group4 = case_when(
    age_class %in%
      c("5-12", "13-15", "16-17") ~ "Youth",
    age_class %in%
      c("18-19", "20-23", "24-34", "35-39") ~ "Young Adult",
    age_class %in%
      c("40-44", "45-49", "50-54", "55-59", "60-64") ~ "Middle Age",
    age_class %in%
      c("65-69", "70-74", "75-79", "80-999") ~ "Senior",
    TRUE ~ NA_character_ ))
```

```
#create a side by side plot of deadlift vs bodyweight by age class
```

```
library(patchwork)
```

```
library(ggplot2)
```

```
# According to the age for the Deadlift and the Squats:
```

```
#Deadlift
```

```
age_deadlift <-
  ggplot(df_noempty, aes(x = log(bodyweight_kg), y = log(best3deadlift_kg))) +
  geom_point(aes(color = age_group4), alpha = 0.2) +
  geom_smooth(aes(color = age_group4), method = "lm", se = FALSE) +
  labs( title = "Deadlift", x = "log(bodyweight (kg))",
        y = "log(deadlift (kg))",
        color = "log(Age Group)")
```

```
#Squats
```

#Effects of the Age group on Squat is more as we can see that the

```
age_squats <-
  ggplot(df_noempty, aes(x = log(bodyweight_kg), y = log(best3squat_kg))) +
  geom_point(aes(color = age_group4), alpha = 0.2) +
  geom_smooth(aes(color = age_group4), method = "lm", se = FALSE) +
  labs( title = "Squat", x = "log(bodyweight (kg))",
        y = "log(squat(kg))",
        color = "Age Group")

age_deadlift +
  age_squats +
  plot_annotation(title = "Figure 1: Bodyweight and Lifting Performance by Age ")
```

Figure 1: Bodyweight and Lifting Performance by Age



#Same but this time facet it by sex

```
library(patchwork)
library(ggplot2)
```

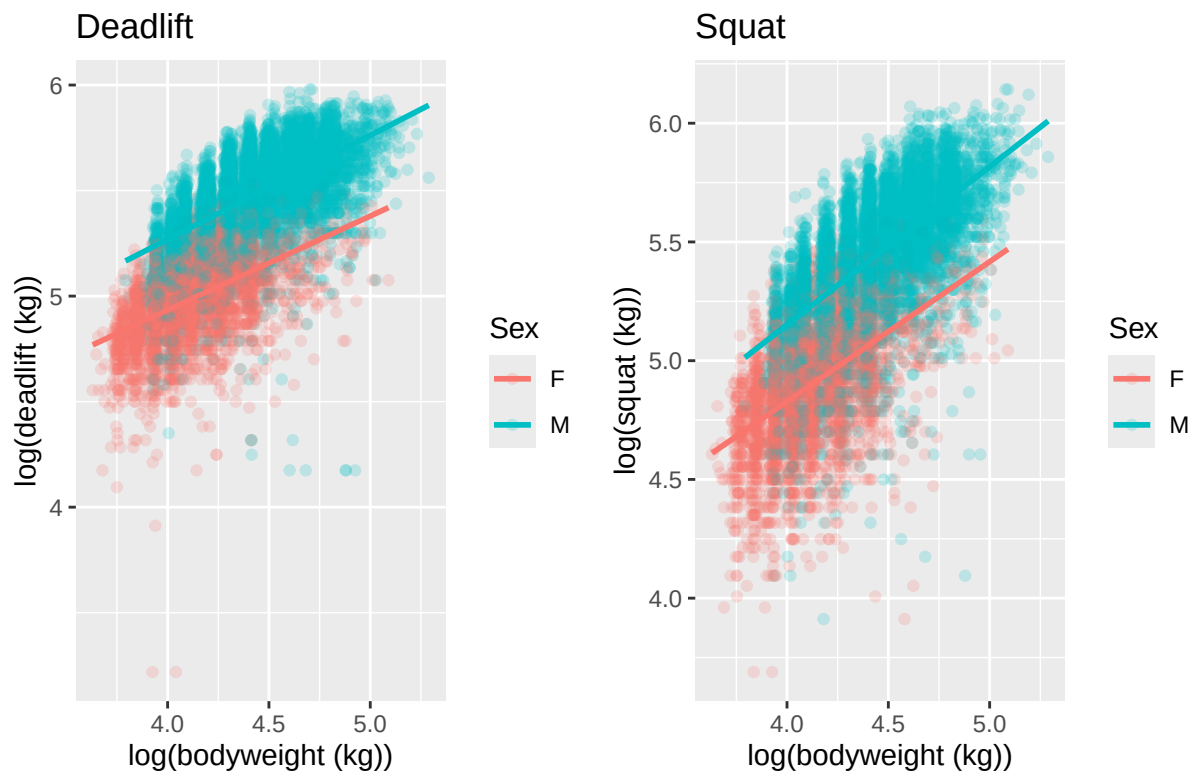
#for the deadlift:

```
sex_deadlift <-
  ggplot(df_noempty, aes(x = log(bodyweight_kg), y = log(best3deadlift_kg))) +
  geom_point(aes(color = sex), alpha = 0.2) +
  geom_smooth(aes(color = sex), method = "lm", se = FALSE) +
  labs( title = "Deadlift", x = "log(bodyweight (kg))",
        y = "log(deadlift (kg))",
        color = "Sex")

#Squat
sex_squats <-
  ggplot(df_noempty, aes(x = log(bodyweight_kg), y = log(best3squat_kg))) +
  geom_point(aes(color = sex), alpha = 0.2) +
  geom_smooth(aes(color = sex), method = "lm", se = FALSE) +
  labs( title = "Squat", x = "log(bodyweight (kg))",
        y = "log(squat (kg))",
        color = "Sex")

sex_deadlift +
  sex_squats +
  plot_annotation(title = "Figure 2: Bodyweight and Lifting Performance by Sex")
```

Figure 2: Bodyweight and Lifting Performance by Sex



```

#create our model for deadlift vs age, sex and bodyweight
full_model_deadlift <-
  lm(log(best3deadlift_kg) ~ log(bodyweight_kg) * sex +
      log(bodyweight_kg) * age_group4, data = df_noempty)

#do the same for squat
full_model_squat <-
  lm(log(best3squat_kg) ~ log(bodyweight_kg) * sex +
      log(bodyweight_kg) * age_group4, data = df_noempty)

# plot the residuals and qq-plot of our model
# in order to determine if a transformation is
#needed and see if any of the assumptions of
#linear regression are violated

library(broom)
augmented_deadlift <-
  augment(full_model_deadlift)

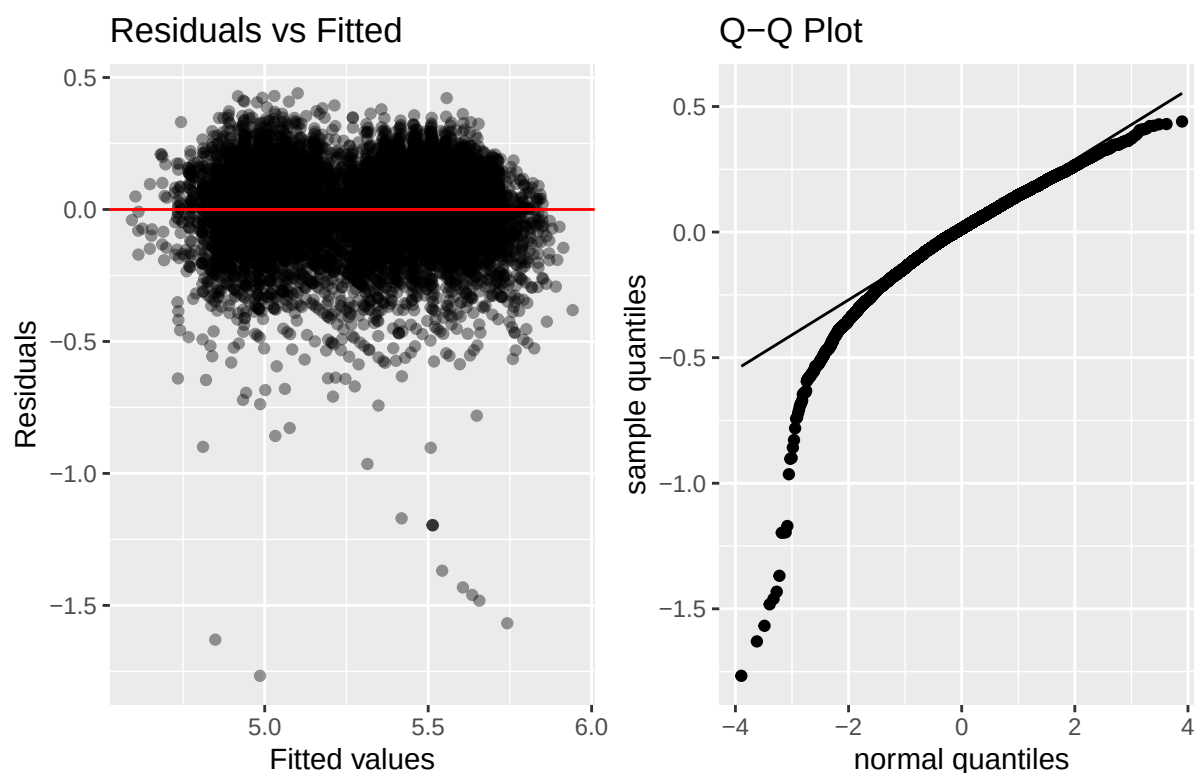
resid <- ggplot(augmented_deadlift, aes(.fitted, .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted", x = "Fitted values", y = "Residuals")

qq <- ggplot(augmented_deadlift, aes(sample = .resid)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "normal quantiles", y = "sample quantiles", title = "Q-Q Plot")

resid +
  qq +
  plot_annotation(title = " Figure 3: Residuals vs Fitted Values and Q-Q Plot")

```

Figure 3: Residuals vs Fitted Values and Q-Q Plot



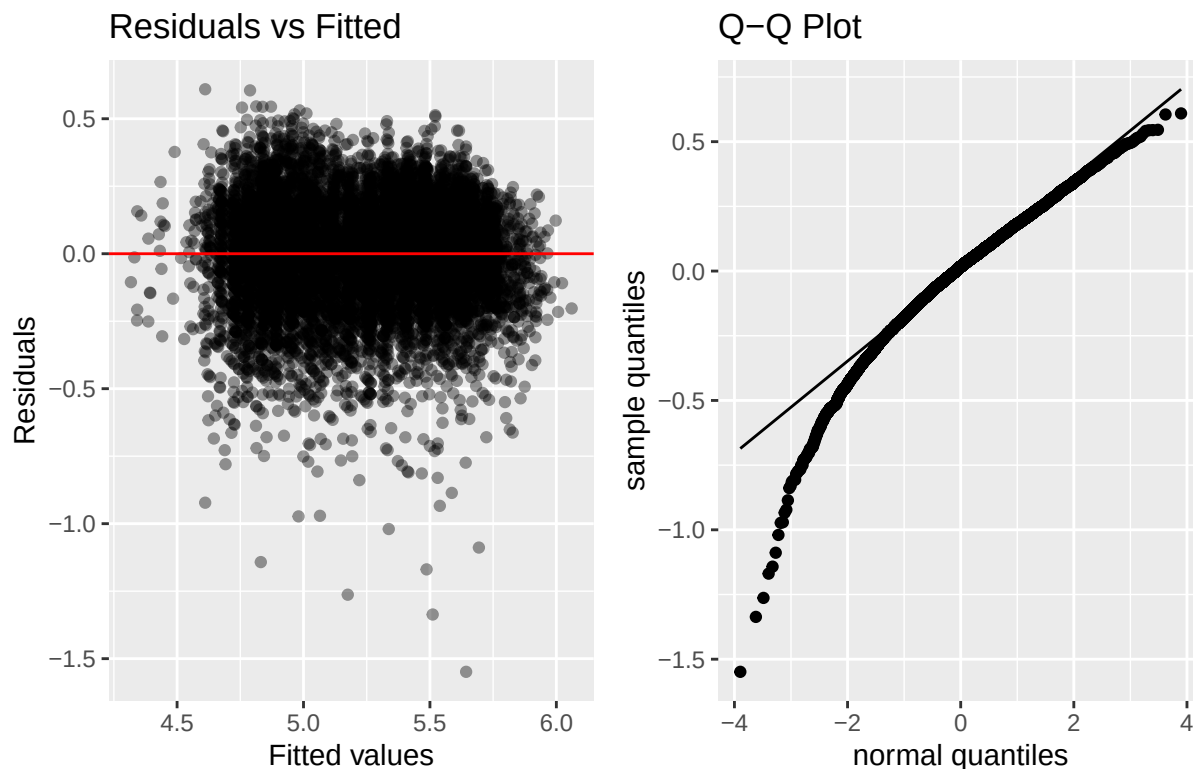
```
#do the same for squat
augmented_squat <- augment(full_model_squat)

resid <- ggplot(augmented_squat, aes(.fitted, .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted", x = "Fitted values", y = "Residuals")

qq <- ggplot(augmented_squat, aes(sample = .resid)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "normal quantiles", y = "sample quantiles", title = "Q-Q Plot")

resid +
  qq +
  plot_annotation(title = " Figure 3: Residuals vs Fitted Values and Q-Q Plot")
```

Figure 3: Residuals vs Fitted Values and Q-Q Plot



```
# plot all of the case influence stats in order to see
# if there are outliers
# and if they are significantly affecting our model
cooks_d_deadlift <-
  ggplot(augmented_deadlift, aes( x = 1:nrow(augmented_deadlift), y = .cooks_d)) +
  geom_point() +
  labs(y = "Cook's Distance", x = "Index") +
  geom_hline(yintercept = 1, linetype = "dashed")

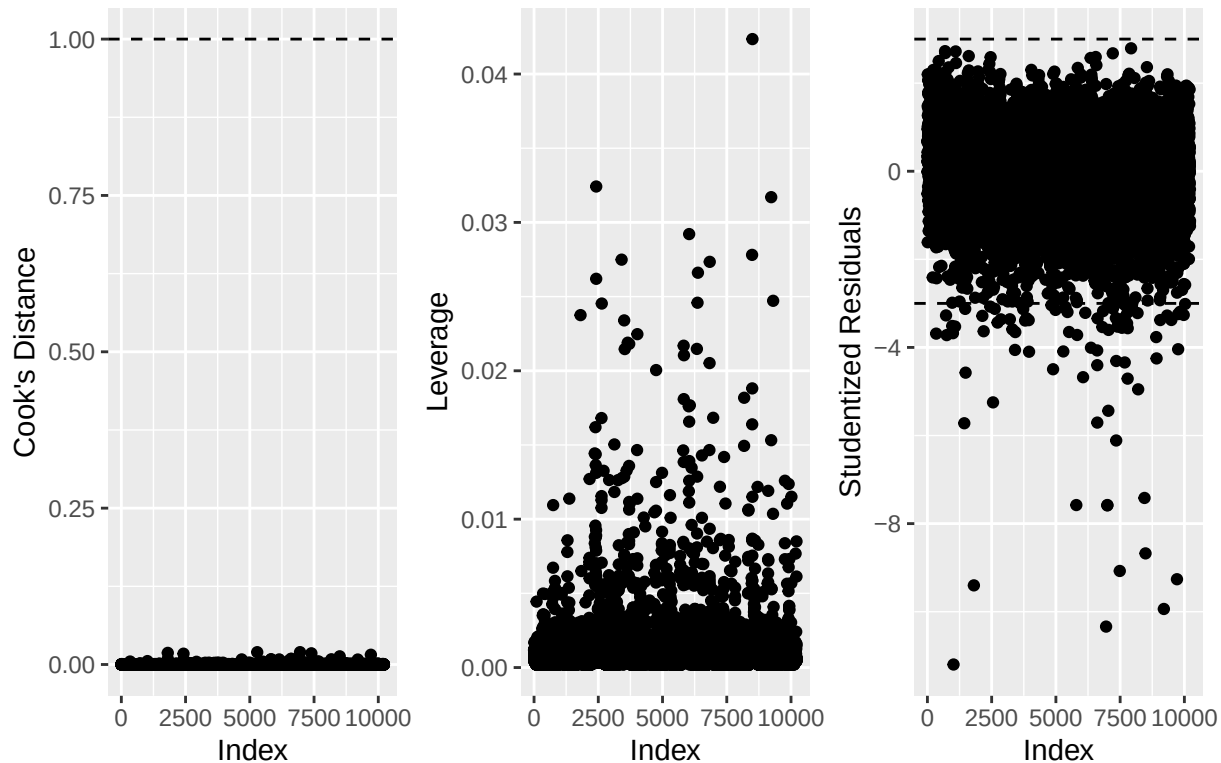
leverage_deadlift <-
  ggplot(augmented_deadlift, aes( x = 1:nrow(augmented_deadlift), y = .hat)) +
  geom_point() +
  labs(y = "Leverage", x = "Index")

studentized_deadlift <-
  ggplot(augmented_deadlift, aes(x = 1:nrow(augmented_deadlift), y = .std.resid)) +
  geom_point() +
  geom_hline(yintercept = 3, linetype = "dashed" ) +
  geom_hline(yintercept = -3, linetype = "dashed") +
  labs(y = "Studentized Residuals", x = "Index")

cooks_d_deadlift +
```

```
leverage_deadlift +
studentized_deadlift +
plot_annotation(title = " Figure 4: Diagnostic Plots for Influential Points")
```

Figure 4: Diagnostic Plots for Influential Points



```
#do the same for squat
cooksd_squat <-
  ggplot(augmented_squat, aes( x = 1:nrow(augmented_squat), y = .cooksd)) +
  geom_point() + labs(y = "Cook's Distance", x = "Index") +
  geom_hline(yintercept = 1, linetype = "dashed")

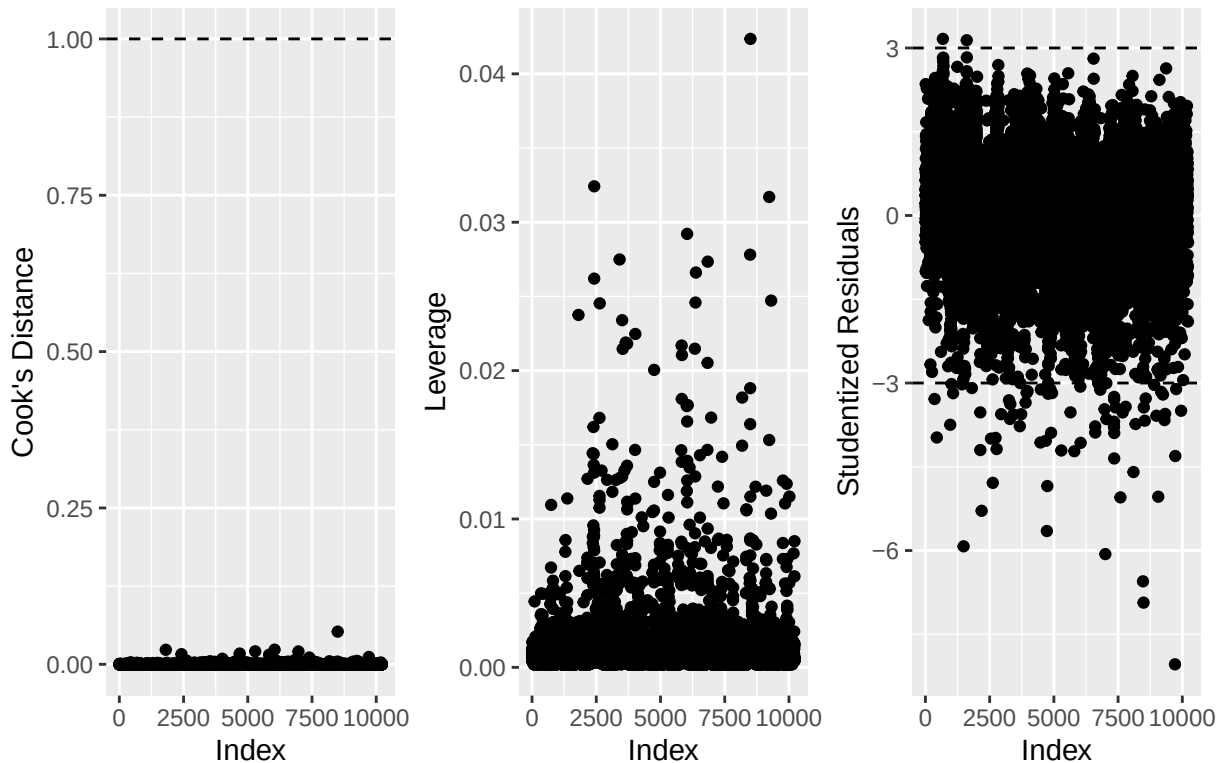
leverage_squat <-
  ggplot(augmented_squat, aes( x = 1:nrow(augmented_squat), y = .hat)) +
  geom_point() +
  labs(y = "Leverage", x = "Index")

studentized_squat <-
  ggplot(augmented_squat, aes(x = 1:nrow(augmented_squat), y = .std.resid)) +
  geom_point() +
  geom_hline(yintercept = 3, linetype = "dashed" ) +
  geom_hline(yintercept = -3, linetype = "dashed") +
  labs(y = "Studentized Residuals", x = "Index")

cooksd_squat +
```

```
leverage_squat +
studentized_squat +
plot_annotation(title = "Figure 5: Diagnostic Plots for Influential Points")
```

Figure 5: Diagnostic Plots for Influential Points



```
#doing an ANOVA F-test for all types of nested models in order to determine
#whether the full interaction model is valid
small_model_deadlift1 <-
  lm(log(best3deadlift_kg) ~ log(bodyweight_kg) * age_group4 +
      sex, data = df_noempty)
anova(small_model_deadlift1, full_model_deadlift)
```

Analysis of Variance Table

```
Model 1: log(best3deadlift_kg) ~ log(bodyweight_kg) * age_group4 + sex
Model 2: log(best3deadlift_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) *
  age_group4
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10210	254.38				
2	10209	254.24	1	0.13556	5.4433	0.01966 *

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
small_model_deadlift2 <-
  lm(log(best3deadlift_kg) ~ log(bodyweight_kg) +
      sex +
      age_group4, data = df_noempty)
anova(small_model_deadlift2, full_model_deadlift)
```

Analysis of Variance Table

```
Model 1: log(best3deadlift_kg) ~ log(bodyweight_kg) + sex + age_group4
Model 2: log(best3deadlift_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) *
  age_group4
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1   10213 254.60
2   10209 254.24  4    0.35677 3.5815 0.006348 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
small_model_deadlift3 <-
  lm(log(best3deadlift_kg) ~ log(bodyweight_kg) * sex +
      age_group4, data = df_noempty)
anova(small_model_deadlift3, full_model_deadlift)
```

Analysis of Variance Table

```
Model 1: log(best3deadlift_kg) ~ log(bodyweight_kg) * sex + age_group4
Model 2: log(best3deadlift_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) *
  age_group4
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1   10212 254.44
2   10209 254.24  3    0.19997 2.6766 0.04546 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#compare the ANOVA F-test to the the p-values in the summary table
summary(full_model_deadlift)
```

Call:

```
lm(formula = log(best3deadlift_kg) ~ log(bodyweight_kg) * sex +
    log(bodyweight_kg) * age_group4, data = df_noempty)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.76682	-0.08495	0.01363	0.10325	0.44031

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.149170	0.068173	46.194	< 2e-16
log(bodyweight_kg)	0.432945	0.016012	27.039	< 2e-16
sexM	0.214729	0.058953	3.642	0.000271
age_group4Senior	-0.218105	0.237687	-0.918	0.358841
age_group4Young Adult	-0.008496	0.058485	-0.145	0.884496
age_group4Youth	0.072999	0.102715	0.711	0.477288
log(bodyweight_kg):sexM	0.032480	0.013922	2.333	0.019663
log(bodyweight_kg):age_group4Senior	-0.004964	0.054759	-0.091	0.927777
log(bodyweight_kg):age_group4Young Adult	0.023655	0.013357	1.771	0.076586
log(bodyweight_kg):age_group4Youth	-0.029583	0.024142	-1.225	0.220464

```

(Intercept)          ***
log(bodyweight_kg)    ***
sexM                  ***
age_group4Senior
age_group4Young Adult
age_group4Youth
log(bodyweight_kg):sexM      *
log(bodyweight_kg):age_group4Senior
log(bodyweight_kg):age_group4Young Adult .
log(bodyweight_kg):age_group4Youth
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1578 on 10209 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.7419, Adjusted R-squared: 0.7417

F-statistic: 3261 on 9 and 10209 DF, p-value: < 2.2e-16

```

#do the same for squat
small_model_squat1 <-
  lm(log(best3squat_kg) ~ log(bodyweight_kg) * age_group4 +
      sex, data = df_noempty)
anova(small_model_squat1, full_model_squat)
```

Analysis of Variance Table

Model 1: log(best3squat_kg) ~ log(bodyweight_kg) * age_group4 + sex

Model 2: log(best3squat_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) * age_group4

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10212	380.20				
2	10211	379.82	1	0.3822	10.275	0.001352 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
small_model_squat2 <-  
  lm(log(best3squat_kg) ~ log(bodyweight_kg) +  
      sex +  
      age_group4, data = df_noempty)  
anova(small_model_squat2, full_model_squat)
```

Analysis of Variance Table

Model 1: log(best3squat_kg) ~ log(bodyweight_kg) + sex + age_group4

Model 2: log(best3squat_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) *
age_group4

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10215	380.52				
2	10211	379.82	4	0.70013	4.7056	0.0008576 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
small_model_squat3 <-  
  lm(log(best3squat_kg) ~ log(bodyweight_kg) * sex +  
      age_group4, data = df_noempty)  
anova(small_model_squat3, full_model_squat)
```

Analysis of Variance Table

Model 1: log(best3squat_kg) ~ log(bodyweight_kg) * sex + age_group4

Model 2: log(best3squat_kg) ~ log(bodyweight_kg) * sex + log(bodyweight_kg) *
age_group4

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10214	380.05				
2	10211	379.82	3	0.23264	2.0847	0.09995 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary(full_model_squat)
```

Call:

```
lm(formula = log(best3squat_kg) ~ log(bodyweight_kg) * sex +  
    log(bodyweight_kg) * age_group4, data = df_noempty)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.54879	-0.11178	0.01676	0.12857	0.60917

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.25008	0.08332	27.007	< 2e-16
log(bodyweight_kg)	0.61555	0.01957	31.456	< 2e-16
sexM	0.12373	0.07204	1.717	0.08594
age_group4Senior	-0.02490	0.29049	-0.086	0.93170
age_group4Young Adult	0.22523	0.07147	3.151	0.00163
age_group4Youth	0.32061	0.12553	2.554	0.01066
log(bodyweight_kg):sexM	0.05454	0.01701	3.205	0.00135
log(bodyweight_kg):age_group4Senior	-0.07700	0.06692	-1.151	0.24996
log(bodyweight_kg):age_group4Young Adult	-0.01556	0.01632	-0.953	0.34050
log(bodyweight_kg):age_group4Youth	-0.06835	0.02950	-2.317	0.02054

(Intercept)	***
log(bodyweight_kg)	***
sexM	.
age_group4Senior	
age_group4Young Adult	**
age_group4Youth	*
log(bodyweight_kg):sexM	**
log(bodyweight_kg):age_group4Senior	
log(bodyweight_kg):age_group4Young Adult	
log(bodyweight_kg):age_group4Youth	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1929 on 10211 degrees of freedom

Multiple R-squared: 0.7241, Adjusted R-squared: 0.7238

F-statistic: 2977 on 9 and 10211 DF, p-value: < 2.2e-16

#calculate AIC for all nested models and full interaction model to support findings in ANOVA

```
AIC(small_model_squat3, small_model_squat2, small_model_squat1,
    full_model_squat)
```

	df	AIC
small_model_squat3	8	-4624.545
small_model_squat2	7	-4613.980
small_model_squat1	10	-4616.524
full_model_squat	11	-4624.804

#do the same for squat

```
AIC(small_model_deadlift3, small_model_deadlift2, small_model_deadlift1,
    full_model_deadlift)
```

	df	AIC
small_model_deadlift3	8	-8721.824

```

small_model_deadlift2 7 -8717.529
small_model_deadlift1 10 -8720.411
full_model_deadlift 11 -8723.858

# Making a table for the deadlift and squat performance increase per doubling
# of bodyweight for each type of lifter (sex × age group)

library(dplyr)
library(tibble)

# Base effects
bd1 <- 0.4329 # Deadlift
bsq <- 0.61555 # Squat

# Coefficients for the Deadlift
mdl <- 0.0325
sdl <- -0.0050
yadl <- 0.0237
ythdl <- -0.0296

# Coefficients for the Squat
msq <- 0.0545
ssq <- -0.0770
yadsq <- -0.0156
ythsq <- -0.0684

table <- tribble(
  ~Group, ~Deadlift_exp, ~Deadlift_Percentage, ~Squat_exp, ~Squat_Percentage,

  "Female, Mid-age",
    bd1, (2^bd1 - 1) * 100,
    bsq, (2^bsq - 1) * 100,

  "Female, Youth",
    bd1 + ythdl, (2^(bd1 + ythdl) - 1) * 100,
    bsq + ythsq, (2^(bsq + ythsq) - 1) * 100,

  "Female, Young Adult",
    bd1 + yadl, (2^(bd1 + yadl) - 1) * 100,
    bsq + yadsq, (2^(bsq + yadsq) - 1) * 100,

  "Female, Senior",
    bd1 + sdl, (2^(bd1 + sdl) - 1) * 100,
    bsq + ssq, (2^(bsq + ssq) - 1) * 100,

```

```

"Male, Mid-age",
  bdl + mdl, (2^(bdl + mdl) - 1) * 100,
  bsq + msq, (2^(bsq + msq) - 1) * 100,

"Male, Youth",
  bdl + mdl + ythdl, (2^(bdl + mdl + ythdl) - 1) * 100,
  bsq + msq + ythsq, (2^(bsq + msq + ythsq) - 1) * 100,

"Male, Young Adult",
  bdl + mdl + yadl, (2^(bdl + mdl + yadl) - 1) * 100,
  bsq + msq + yadsq, (2^(bsq + msq + yadsq) - 1) * 100,

"Male, Senior",
  bdl + mdl + sdl, (2^(bdl + mdl + sdl) - 1) * 100,
  bsq + msq + ssq, (2^(bsq + msq + ssq) - 1) * 100
)

table

# A tibble: 8 x 5
  Group                Deadlift_exp Deadlift_Percentage Squat_exp Squat_Percentage
  <chr>                <dbl>             <dbl>         <dbl>         <dbl>
1 Female, Mid-age      0.433              35.0          0.616          53.2
2 Female, Youth        0.403              32.3          0.547          46.1
3 Female, Young Adu~   0.457              37.2          0.600          51.6
4 Female, Senior       0.428              34.5          0.539          45.3
5 Male, Mid-age        0.465              38.1          0.670          59.1
6 Male, Youth          0.436              35.3          0.602          51.7
7 Male, Young Adult    0.489              40.4          0.654          57.4
8 Male, Senior         0.460              37.6          0.593          50.8

# Making a table of the coefficients of the interaction terms in both models
# to interpret their values (as percentage change in performance per doubling
# of bodyweight)

# Deadlift coefficients
dl_m  <- 0.032480
dl_s  <- -0.004964
dl_ya <- 0.023655
dl_y  <- -0.029583

# Squat coefficients
sq_m  <- 0.05454
sq_s  <- -0.07700
sq_ya <- -0.01556

```

```
sq_y <- -0.06835

effects <- tribble(
  ~Group, ~DL_Percent, ~SQ_Percent,

  "Male vs Female",
    (2^dl_m - 1) * 100,    (2^sq_m - 1) * 100,

  "Senior vs Mid-age",
    (2^dl_s - 1) * 100,    (2^sq_s - 1) * 100,

  "Young Adult vs Mid-age",
    (2^dl_ya - 1) * 100,   (2^sq_ya - 1) * 100,

  "Youth vs Mid-age",
    (2^dl_y - 1) * 100,    (2^sq_y - 1) * 100
)

effects
```

```
# A tibble: 4 x 3
  Group          DL_Percent SQ_Percent
  <chr>          <dbl>     <dbl>
1 Male vs Female      2.28       3.85
2 Senior vs Mid-age  -0.343     -5.20
3 Young Adult vs Mid-age  1.65     -1.07
4 Youth vs Mid-age    -2.03     -4.63
```